Asexual Reproduction

Exam code: 0610 & 0970

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Asexual Reproduction

- Asexual reproduction does not involve sex cells or fertilisation
- **Only one parent is required** so there is no fusion of gametes and no mixing of genetic information
- As a result, the offspring are **genetically identical to the parent and to each other** (clones)
- Asexual reproduction is defined as **a process resulting in genetically identical offspring from one parent**

Examples of Asexual Reproduction

Bacteria produce exact genetic copies of themselves in a type of asexual reproduction called binary fission:

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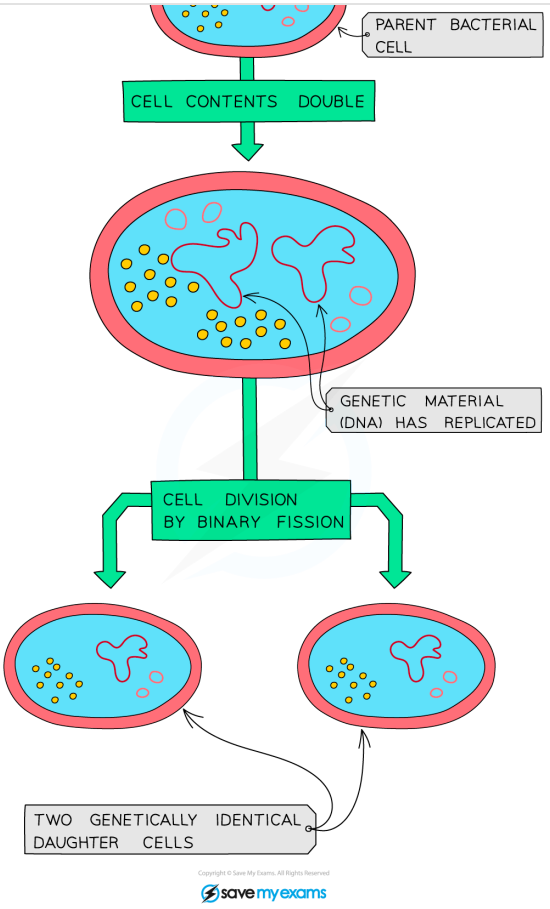

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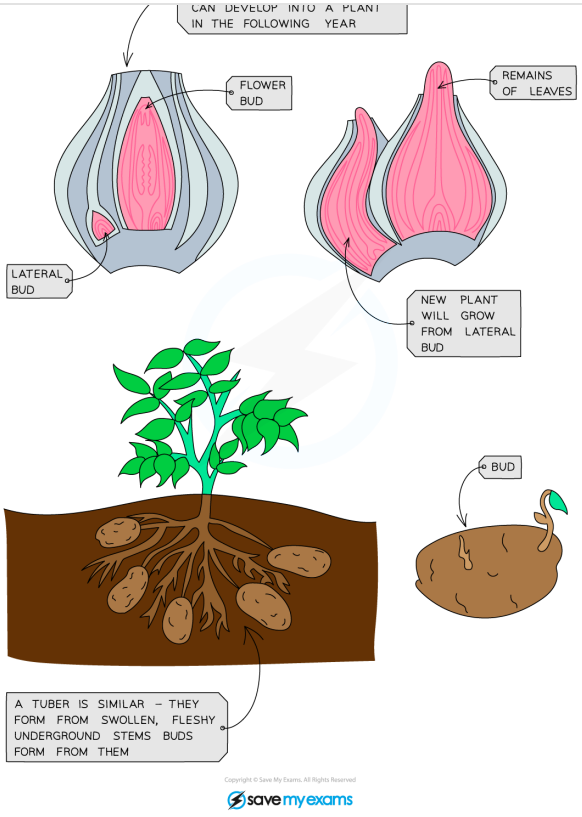
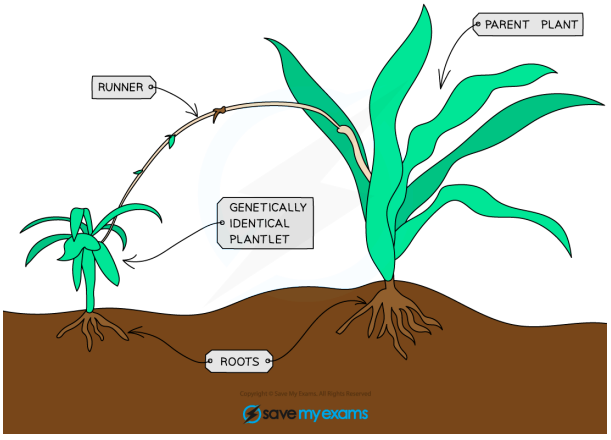
Plants can reproduce asexually using bulbs and tubers; these are food storage organs from which budding can occur, producing new plants which are genetically identical to the parent plant:



Some plants develop underground food storage organs that will develop into next years plants – they can take different forms, such as bulbs or tubers

Some plants grow side shoots called runners that contain tiny plantlets on them (a good example of this are strawberry plants. These will grow roots and develop into separate plants, again being genetically identical to the parent plant:



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Disadvantages of Asexual Reproduction: Extended

Species Type	Advantages	Disadvantages
Wild species	Rapid population growth	Limited genetic variation in population - offspring are genetically identical to their parents
	Can exploit suitable environments quickly	Vulnerability to habitat changes e.g. temperature changes, droughts or new predators
	More time and energy-efficient	Disease is likely to affect the whole population as there is no genetic variation
	Reproduction is completed much faster than sexual reproduction	No recombination of genes means evolution is slow
Crop plants	Crops can be produced with desired characteristics e.g. high yield, disease-resistant, drought resistant	Crops will not adapt to changes in climate if produced asexually
	Crops can be produced with uniform characteristics necessary for commercial sale	The process of asexual reproduction in crops requires human input and management
	Production of crops can be faster	If a diseased parent plant is used in the reproduction process, the offspring will also be diseased
	The cost of production will be less than investing in seeds	Vulnerability to disease or pests may result in big financial losses for the farmer

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